RAPIDSEGMENT

How the Engine Discovers High-Lift Hierarchical Segments

A walkthrough of end-to-end rule mining pipeline.

Technical briefing

AGENDA

What we will cover today

01 **What RapidSegment is**

Purpose and positioning vs black-box models and decision trees

02 **End-to-end data flow**

From raw table to hierarchical segments and production SQL

03 **Binning & variable selection**

How features are ranked and discretized (Optimal vs Naive)

04 **Combinations & constraints**

How 1-/2-/3-way rules are generated, pruned, and filtered

05 **Ranking & hierarchy**

Champion selection, residual update, final ordered segments

06 **Outputs & control levers**

SQL filters, scorecard, and the parameters that matter

WHAT IT IS

Transparent segments that maximise lift over baseline

In one sentence

RapidSegment is a combinatorial heuristic engine that discovers high-lift predictive segments and turns them into production-ready SQL rules and scorecards.

It is designed for

- Credit risk, collections, marketing response, fraud flags
- Teams that need explainable rules, not black boxes
- Direct deployment into SQL / policy engines / scorecards

Vs Decision Trees / ML

Goal
High-lift, business-ready segments

Shape
Explicit hierarchical rules + SQL

Reuse
Controlled feature reuse across segments

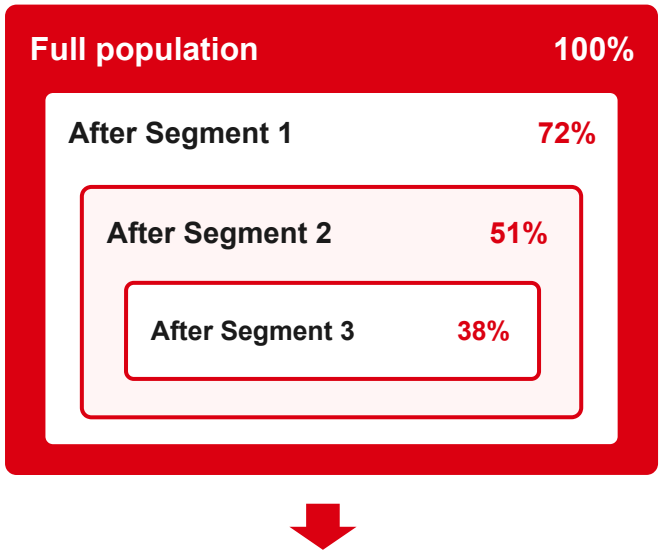
Output
SQL filters + integer weights

Ops fit
Policy, scorecards, rules engines

From full data to hierarchical segments & scores

RESIDUAL DATA

Starts full · shrinks after each segment



01 Residual data
Start with full table; shrink after each segment is accepted

02 IV + Binning
Rank features by IV (or response rate), create discrete bins

03 Combinations
1-, 2-, 3-way rules with Apriori pruning & diversity checks

04 Hard filters
min_sample, min_lift, min_events must pass on residual data

05 Champion
Rank by sort_priority (rate x lift x count) pick best rule

06 Remove & repeat
Delete matching rows; next segment is next priority

07 Score & decile
Flags -> weights (RR x 100) -> score; deciles on score > 0

FINAL HIERARCHY

(priority order = discovery order)

Seg 1
max_dpd >= 60 AND util >= 0.85
Highest priority

>

Seg 2
risk_band = High
Applies to remainder

>

Seg 3
util >= 0.90
Applies to remainder

>

Else
Everyone else
Baseline

KEY IDEA

Find the best rule on what's left -> lock it in -> remove those customers -> repeat.

Then score the full dataset: segment flags x weights (response rate x 100) -> customer score -> deciles on score > 0 only.

Order of discovery becomes priority in the hierarchy (SQL CASE WHEN). Scoring is independent of residual order.

Binning -> Combinations -> Hard constraints -> Ranking -> Residual shrink -> Hierarchy -> Score & decile

Turn raw features into ranked, discrete building blocks

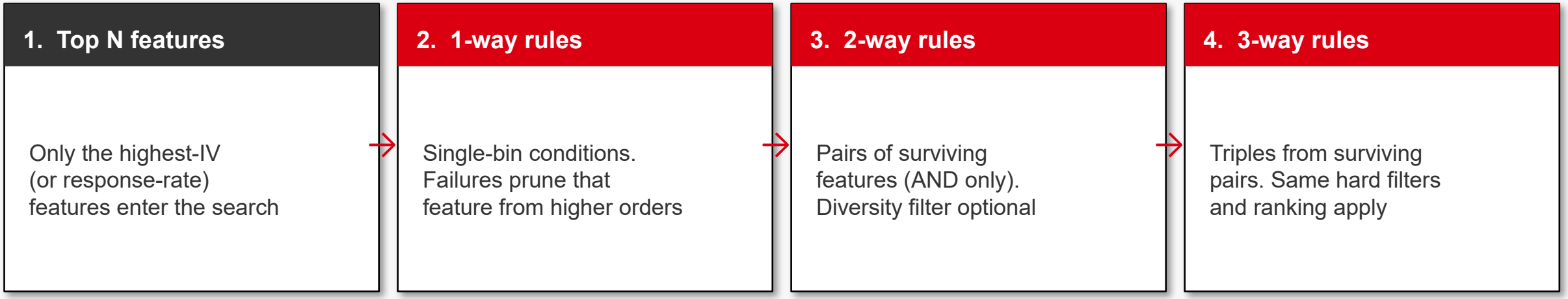
Optimal Binning (default)

Engine	optbinning (CART or Quantile pre-binning)
Metric	Information Value (IV) or Response Rate per feature
Numeric	Numeric bins are auto merged to optimize IV
Categorical	Categories are auto merged to optimize IV
Strength	Target-aware cuts → usually higher lift
When	Accuracy of cuts matters more than speed

Naive Binning (fast path)

Engine	Pure DuckDB SQL (QUANTILE_CONT)
Metric	Information Value (IV) or Response Rate per feature
Numeric	Equal-frequency quantiles (5,10,20…)
Categorical	One bin per distinct value + Missing
Strength	Very fast; supports adjacent-bin expansion
When	Large data or stable, interpretable bins

Apriori-style search: only successful lower-order rules grow



Apriori-style pruning working example

Design choices that keep search trackable and business-friendly

AND only

OR would break Apriori pruning and explode the search space

Feature groups

Optional diversity: block rules that mix variables from the same business group

max_feature_reuse

Caps how many times one feature can appear across the final segment list

Expansion hops

Optional merge of neighbouring bins to recover higher-event rules

HARD CONSTRAINTS

Every candidate must clear absolute floors before ranking

min_sample_size

Minimum row count in the residual that the rule must cover. Protects against tiny, unstable cells.

min_lift

Rule response rate must be at least this multiple of the current residual base rate.

min_events

Minimum number of positive target events inside the rule. Ensures statistical credibility.

Feature reuse / diversity

max_feature_reuse and optional group diversity stop the same variables dominating every segment.

sort_priority decides which surviving rule wins each iteration

Default and common ranking keys (first dimension is primary)

`rate_lift_count`	Default — prefer highest response rate, then lift, then volume
`lift_count_rate`	Classic lift-first view of portfolio impact
`count_lift_rate`	Volume-first when coverage is the business priority
`events_lift_rate`	Event-count first (good for rare-event targets)
`rate_events_lift`	High purity + event mass

Grid search (optional)

When param_grid is supplied, the engine evaluates several (min_sample_size, min_lift) pairs per iteration. Each pair produces a candidate; the global champion is still selected by the same sort_priority. This lets the model adapt thresholds as the residual base rate and volume change.

FINAL HIERARCHICAL SEGMENTS

Order of discovery = priority order in the final assignment

1. **Accept champion**
   Rule is stored with SQL filter, count, rate, lift measured on residual
2. **Remove matches**
   Residual updated with NULL-safe logic: WHERE NOT (rule) OR rule IS NULL
3. **Next iteration**
   IV, bins, combinations re-computed on the smaller residual
4. **Stop conditions**
   max_segments reached, or no rule clears hard constraints, or volume too low
5. **Final coverage**
   CASE WHEN hierarchy applied back to original data for true counts

SCORING THE FINAL DATASET

Segments → Scores → Deciles

1. **Full dataset**
   Score every row on the original population (not the residual).

2. **Segment flags**
   Each segment SQL runs independently → binary column (0 / 1) per segment.

3. **Segment weights**
   Weight = segment response rate × 100, rounded. Higher rate → higher weight.

4. **Customer score**
   Score = sum of weights for every segment flag that is 1 on that row.

5. **Deciles (score > 0 only)**
   Split active scores into 10 buckets. Zero-score rows are excluded from thresholds.

WORKED EXAMPLE

Segment weights (from response rate)

Segment	Resp. rate	Weight
Seg 1	18%	**18**
Seg 2	12%	**12**
Seg 3	9%	**9**

One customer example

Seg 1 flag	**1**	+ 18
Seg 2 flag	**0**	+ 0
Seg 3 flag	**1**	+ 9

Total score = 18 + 0 + 9 = 27

Deciles are cut only on customers with score > 0.
Score = 0 stays in the baseline / inactive pool.

WHAT YOU GET

Production artefacts from a single extract_segments run

SQL filters

Pure ANSI SQL WHERE clauses for every segment. Ready for rules engines, BI tools, or batch scoring.

Hierarchical list

Ordered segments with count, response rate, lift, and meta (thresholds that produced the rule).

Scorecard weights

StrategicSegmentScore converts flags into integer weights from response rate and calibrates deciles.

Full audit trail

explain_feature_journey, explain_no_segments, feature health reports — complete diagnostics.

CONTROL LEVERS

Full Flexibility on Segment Construction & Selection

`top_n_vars`	How many features enter the combinatorial search (default 15)
`max_segments`	Hard cap on the length of the hierarchy
`min_lift / min_sample_size / min_events`	Absolute quality floors — the “hard constraints”
`param_grid`	Sweep alternative floors; champion still chosen by sort_priority
`sort_priority`	Business preference: rate-first, lift-first, volume-first, events-first
`binning_method`	optimal_cart / optimal_quantile / naive
`max_feature_reuse`	Prevent one variable from owning every segment
`enable_diversity + feature_groups`	Force cross-group combinations only
`max_expansion_hops`	Allow neighbouring bins to merge (0 = off)

SUMMARY

Five things to remember

1. Binning + IV/RR selects a shortlist of predictive features and turns them into discrete bins.
2. Combinations are grown bottom-up (1 → 2 → 3 way) with Apriori pruning and optional diversity.
3. Hard constraints (sample, lift, events) are absolute gates — ranking only happens among survivors.
4. The champion of each residual is kept; matching rows are removed; the next champion is the next priority.
5. Output is hierarchical SQL rules + optional scorecard weights — ready for production and audit.

How 1-way, 2-way and 3-way rules are built (dummy data)

Dummy residual · Base rate = 5% | Features ranked by IV: max_dpd · util · risk_band · tenure · product

Hard floors for this example: min_sample = 1,000 · min_lift = 1.5× · min_events = 50

1-WAY

Single-bin conditions

- ✓ **max_dpd ≥ 60**
  n=2,400 rate=12% lift=2.4× ✓
- ✓ **util ≥ 0.85**
  n=1,800 rate=9% lift=1.8× ✓
- ✓ **risk_band = High**
  n=3,100 rate=8% lift=1.6× ✓
- ✗ **tenure < 6m**
  n=900 rate=6% lift=1.2× ✗ sample
- ✗ **product = Gold**
  n=4,000 rate=5.2% lift=1.04× ✗ lift

Only features that pass 1-way enter 2-way search.

2-WAY

AND of two surviving features

- ✓ **max_dpd≥60 AND util≥0.85**
  n=1,200 rate=18% lift=3.6× ✓
- ✓ **max_dpd≥60 AND risk=High**
  n=1,500 rate=14% lift=2.8× ✓
- ✓ **util≥0.85 AND risk=High**
  n=1,100 rate=11% lift=2.2× ✓
- ✗ **… any pair with tenure**
  tenure failed 1-way → pruned
- ✗ **… any pair with product**
  product failed 1-way → pruned

Apriori: failed 1-way features are never combined further.

3-WAY

AND of three surviving features

- ✗ **max_dpd≥60 ∧ util≥0.85 ∧ risk=High**
  n=850 rate=22% lift=4.4× ✗ sample
- ✗ **(no other triple clears floors)**
  All other triples fail min_sample or min_events

In this run the 3-way fails min_sample → only 1- & 2-way candidates go to ranking.

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